

EDUCATION

University of Rochester

Expected June 2027

B.S. Statistics & B.S. Finance | GPA: 3.70 / 4.00 | Dean's List All Semesters

Rochester, NY

Certificate of Achievement in Actuarial Studies (In Progress)

Relevant Coursework: Statistical Methodology, Financial Statement Analysis, Investments, Statistical Computing in R, Financial Management, Accounting, Corporate Finance

Skills: Python, R, SQL, Excel (Advanced), Tableau, Power BI, Financial Modeling, DCF/WACC/CAPM, Sensitivity Analysis, Regression Modeling, Bayesian/MCMC Methods, Monte Carlo Simulation, Data Visualization

Language: English, Mandarin, Cantonese, Fuzhounese

PROJECTS

Beyond U-3: Labor Market Dashboard

May 2026 – Current

Python, PostgreSQL, SQL, Tableau, FRED API, pandas, SQLAlchemy

- Designed and built a normalized PostgreSQL relational database from scratch to store and analyze 14 Federal Reserve economic time series spanning 30+ years of U.S. labor market data
- Wrote advanced SQL queries including multi-table joins, window functions, rolling averages, and CASE-based recession period classification; quantified that the official unemployment rate understates labor stress by an average of 4.42 percentage points in normal periods and 5.37 points during the Great Recession
- Built an automated Python data pipeline using the FRED API, pandas, and SQLAlchemy to programmatically fetch, resample, and load mixed-frequency time series (monthly, weekly, daily) into PostgreSQL; pipeline dynamically calculates derived metrics including U-3/U-6 gap and temp help employment decline from all-time peak
- Scheduled pipeline via Windows Task Scheduler to auto-execute monthly following BLS data releases, maintaining a live production database without manual intervention
- Tracked 8 leading recession indicators including temporary help employment (down 21.4% from peak), quits rate, yield curve spread, personal saving rate, and credit card delinquency as early warning signals
- Built a 4-chart Tableau Public dashboard with intentional axis methodology, synchronized dual axes, and recession-period analysis; dashboard is publicly accessible and linked on GitHub

Quantitative Risk Modeling: S&P 500 Volatility Prediction

January – May 2026

Python, R | Independent Research Project | github.com/marksguo/SPY-Volatility-Modeling

- Built and compared OLS, WLS, and Robust M-estimation models on 4,800+ daily S&P 500 returns (2007–2026) to predict financial volatility and quantify risk persistence.
- Confirmed statistically significant volatility clustering ($p < 0.0001$); findings directly support dynamic VaR estimation and portfolio risk frameworks used in industry.
- Validated model performance via 70/30 train-test split with full diagnostic suite: heteroscedasticity testing, ACF analysis, and Cook's distance influence detection.

Bayesian Hierarchical Modeling: Uncertainty Quantification

September 2025 – March 2026

R | Independent Research Project

- Independently implemented a hierarchical Bayesian model with a custom Metropolis-Hastings sampler; ran 4 Markov chains of 100,000 iterations each to estimate group-level probabilities.
- Confirmed full convergence via Gelman-Rubin diagnostics (PSRF = 1.0); achieved effective sample sizes exceeding 26,000 across all chains.
- Delivered probabilistic findings with 95% credible intervals and actionable business recommendations for non-technical audiences.

Amazon.com, Inc. Financial Statement Analysis & Equity Valuation

January – April 2025

Excel | Independent Research Project

- Built a 5-year pro forma model and DCF valuation using CAPM-derived WACC across multiple macroeconomic scenarios; stress-tested assumptions via sensitivity analysis on beta, WACC, and terminal growth rate.
- Conducted trend, common-size, and ratio analysis across profitability, liquidity, leverage, and default risk; produced a structured equity research report with a supported “Hold” recommendation.

University Financial Position & Resource Allocation Research

February – April 2024

Excel | Collaborative Research with Prof. Michael Rizzo

- Analyzed financial statements and endowment data across 11 universities using NACUBO and IPEDS; identified significant correlation between endowment-per-student and U.S. News rankings.
- Presented findings to Economics faculty with data-driven conclusions on institutional resource allocation tradeoffs.

WORK EXPERIENCE

China Fun of NC, Inc.

September 2017 – July 2025

Junior Financial & Operations Analyst

Charlotte, NC

- Drove 25% revenue growth (~\$375K incremental annually) at a \$1.5M+ revenue operation by identifying margin expansion opportunities and implementing data-driven process improvements across procurement, fulfillment, and labor allocation.
- Monitored and analyzed food cost ratios against a ~32% COGS target; identified \$10K+ in annual ingredient savings through vendor renegotiation and waste reduction strategies across \$30K+ in monthly procurement.
- Built a 2,000-cell Excel order management and financial tracking system integrated with DoorDash, improving transaction accuracy across 150+ daily orders and reducing fulfillment errors — equivalent to reconciling \$125K+ in monthly revenue.
- Prepared monthly P&L summaries tracking revenue, labor cost (~28% of sales), and food cost across dine-in and delivery channels; presented findings to ownership to support operational and budgetary decisions.
- Managed seasonal cash flow planning around high-volume periods (Chinese New Year, summer), maintaining procurement schedules and cash reserves during slow months to sustain consistent operations.
- Reconciled daily POS receipts against bank deposits with <0.5% discrepancy; maintained accounts payable records across multiple suppliers and supported quarterly tax documentation for CPA-led filings.
- Monitored front-of-house vs. back-of-house labor costs as a % of revenue; processed bi-weekly payroll for 15+ employees and ensured tip reporting compliance with IRS guidelines.
- Trained 10+ staff on POS and operational systems, achieving a 30% efficiency gain, leading to reducing per-transaction labor cost and improving throughput during peak periods.

LEADERSHIP AND COMMUNITY INVOLVEMENT

Tien Ren Cultural & Educational Foundation

Summer 2015 – Current

Lecturer & Event Organizer (2021–Present) | Assistant Lecturer (2019–2021)

Indian Trail, NC

- Delivered 30+ educational sessions to ~1,000 youths (ages 14–20); organized 3 annual volunteer events at Second Food Bank of Metrolina.

INTERESTS

Personal Interests: Quantitative Finance, Risk Management, Actuarial Science, Entrepreneurship, Networking, Chinese Literature & History, Mathematics, Self-Study, Music